

Phase	Module	Session	Hours	Topic	Detailed Content	Hands-on / Tools	GenAI Integration
Phase 1: Foundations	Module 1: AI & GenAI Basics	1.1	2	Introduction to AI in Finance	Narrow AI vs. GenAI, ML Paradigms (Supervised, Unsupervised), Fintech Evolution	Discussion: AI in Banking	Intro to Gemini/ChatGPT
Phase 1: Foundations	Module 1: AI & GenAI Basics	1.2	2	Prompt Engineering for Finance	Context window, few-shot prompting, persona setting for financial analysis	Lab: Summarizing 10-K Filings	ChatGPT, Gemini, Claude
Phase 1: Foundations	Module 2: Productivity Tools	2.1	2	Microsoft Copilot in Finance	Excel automation, automating Pivot tables, formula generation, PPT Pitch decks	Lab: DCF Model Outline in Excel	MS 365 Copilot
Phase 1: Foundations	Module 2: Productivity Tools	2.2	2	AI for Code & Debugging	Zero-code EDA, using LLMs to write Python logic, debugging common errors	Lab: Python Script for Stock Data	ChatGPT Plus / Copilot Chat
Phase 2: Predictive Analytics	Module 3: Python Core	3.1	2	Python Essentials	Data types, Lists, Dictionaries, Logic, Loops for financial iteration	Lab: Interest Calculator Script	Google Colab
Phase 2: Predictive Analytics	Module 3: Python Core	3.2	3	Financial Data Wrangling	Pandas & NumPy: Handling OHLC data, cleaning currency strings, reindexing time-series	Lab: Portfolio Data Cleaning	Pandas, NumPy
Phase 3: Machine Learning	Module 4: Regression (Value)	4.1	3	Linear & Multiple Regression	Beta calculation, CAPM concepts, Ordinary Least Squares (OLS)	Lab: Predicting Stock Valuations	Scikit-Learn
Phase 3: Machine Learning	Module 4: Regression (Value)	4.2	2	Regularization Techniques	Overfitting vs. Underfitting, Lasso & Ridge Regression for macro variables	Lab: Housing Price Prediction	Scikit-Learn
Phase 3: Machine Learning	Module 5: Classification (Risk)	5.1	3	Credit Scoring Models	Logistic Regression, Decision Trees, Random Forests for default risk	Lab: Credit Default Predictor	Scikit-Learn
Phase 3: Machine Learning	Module 5: Classification (Risk)	5.2	3	Fraud Detection & Metrics	Confusion Matrix, Precision-Recall, AUC-ROC, SMOTE for imbalanced data	Lab: Credit Card Fraud Detection	Scikit-Learn
Phase 4: Advanced Topics	Module 6: Forecasting & Seg.	6.1	3	Time Series & Clustering	ARIMA basics, K-Means Clustering for wealth management & customer segmentation	Lab: HNW Client Segmentation	Statsmodels, Scikit-Learn
Phase 4: Advanced Topics	Module 7: Capstone	7.1	3	Final Integrated Project	End-to-End: Data Ingestion -> Cleaning -> ML Model -> Financial Interpretation	Project: Market Sentiment Predictor	Full Stack AI/ML